Title: GPIO Control for Alternating LED Patterns

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Introduction-

Before this lab, it was known that microcontrollers contain General Purpose Input/Output (GPIO) pins which can be programmed by software to control electronic components, such as turning LEDs on and off. The lab is about applying this knowledge to create a specific visual pattern—an alternating blink—between two different-colored LEDs on a development board. The purpose of this lab was to gain practical experience in embedded programming by writing and uploading code that controls GPIO pins to create a continuous, timed, alternating sequence between a red and a blue LED.

Materials-

Apparatus	Expendables & Software
• PC/Laptop	• AHL GEC IDE v4.53
• GPIO Development Board	• Project File
• RGB LED Module	• USB Cable
• Power Supply	• Jumper Wires

Procedure-

The procedure was followed as outlined in the lab handout. The key steps involved:

1. Opening the existing "GPIO-Output" project in the AHL GEC IDE.
2. Navigating to and reviewing the main program file (main.c).
3. Modifying the code within the infinite loop to control the RED and BLUE LEDs.
4. Adding a delay function to create a visible blinking speed.
5. Compiling the code and uploading the resulting .hex file to the STM32 development board.
6. Observing the LED behavior on the connected RGB module to verify correct operation.

Data-

The following behavior was observed after the program was successfully uploaded:

- The Red LED turned on, and the Blue LED was off.
- After a short delay, the Red LED turned off, and the Blue LED turned on.
- This alternating pattern continued indefinitely.

Variables Used in the Project:

- GPIO_PIN_RED: The software reference for the pin connected to the red LED.
- GPIO_PIN_BLUE: The software reference for the pin connected to the blue LED.
- ON / OFF: Constants used to set the GPIO pin state to high or low voltage.
- Delay(500): A function call that pauses program execution for 500 milliseconds.

Calculations-

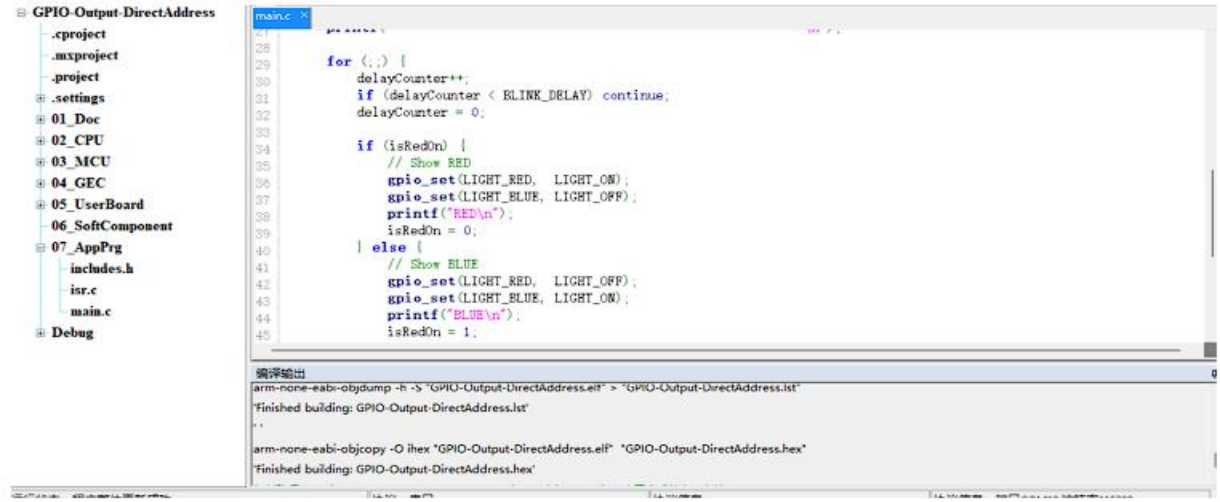
The primary calculation involved determining the blink period. The code alternates the state every 500 ms.

- Time for one full cycle (Red ON → Blue ON → Red ON): 500 ms + 500 ms = 1000 ms
- Blink Frequency: 1 / 1.0 second = 1 Hz

The code snippet below shows how the timing and control logic was implemented:

```
while(1) {
    // Activate RED, deactivate BLUE
    GPIO_SetPin(GPIO_PIN_RED, ON);
    GPIO_SetPin(GPIO_PIN_BLUE, OFF);
    Delay(500); // wait 500 ms

    // Activate BLUE, deactivate RED
    GPIO_SetPin(GPIO_PIN_RED, OFF);
    GPIO_SetPin(GPIO_PIN_BLUE, ON);
    Delay(500); // wait 500 ms
}
```





Analysis

1. What is the role of the Delay() **function in this program?**
The Delay() function pauses the microcontroller's execution for a specified amount of time. Without these delays, the microcontroller would switch the LEDs on and off millions of times per second, which is too fast for the human eye to perceive. The delay creates a pause that makes the alternating pattern visible as a distinct blink.

2. Why is an infinite loop (while(1)**) used in this program?**
An infinite loop is used because the program is intended to run continuously for an indefinite period after the board is powered on. This ensures the alternating LED pattern repeats forever until the board is reset or powered off, which is the desired behavior for many embedded applications like status indicators or warning lights.

Conclusions

The purpose of this lab was to control GPIO pins to create an alternating LED pattern. This was successfully accomplished; the code was written, compiled, and uploaded to the development board, resulting in a red and blue LED blinking on and off in a steady, alternating sequence at a frequency of 1 Hz.

The experiment was successful, but potential sources of error could have included incorrect pin assignments in the code (e.g., confusing the pin for red and blue), syntax errors during code modification, or a faulty physical connection with the jumper wires, which would prevent the electrical signal from reaching the LED.

This exercise relates directly to real-world applications outside the classroom. The same fundamental principles of GPIO control and timing are used in traffic light systems, the blinking status lights on routers and electronics, automotive turn signals, and indicator systems in industrial machinery. Understanding how to program this basic hardware interaction is a foundational skill for a career in electronics or embedded systems engineering.